LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 8 Test (Set B): Heredity

Chapter / Topic: Heredity

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SECTION A

Q1. Plants with round seeds (RR) are crossed with plants with wrinkled seeds (rr). The F_1 seeds are: **[1]**

Answer: (b) all round (Rr)

Q2. How many pairs of chromosomes are there in human body cells? **[1]**

Answer: (c) 23 pairs (46 chromosomes)

Q3. A child has the sex chromosomes XX. The child is: **[1]**

Answer: (d) a girl

Q4. In the F_2 generation of Mendel's dihybrid cross, what fraction of plants show both recessive traits (wrinkled and green)? **[1]**

Answer: (a) 1/16

Q5. Mendel carried out his experiments on: **[1]**

Answer: (b) garden peas

Q6. **Assertion (A):** There is very little variation among offspring produced by asexual reproduction.
Reason (R): In asexual reproduction, small errors in… **[1]**

Answer: (a)

SECTION B

Q7. Why were all the F_1 plants tall, with no medium-height plants, when Mendel crossed pure tall and pure short pea plants? **[2]**

Answer: Tallness (T) is dominant over shortness (t). The F_1 plants are Tt, and a single copy of T is enough to make the plant tall. Traits do not blend, so no medium-height plants appear.

- Tall is dominant, F_1 is Tt: 1
- No blending of traits: 1

OR, answer: Genotype is the genetic make-up of an organism for a trait (e.g. TT, Tt or tt). Phenotype is the visible trait that results (e.g. tall or short).

- Each with example: 1

Q8. A man with blood group A marries a woman with blood group O, and their daughter has blood group O. Is this enough to tell which trait, A or O, is… **[2]**

Answer: No. One child is not enough. If A were dominant, the father could be carrying one O copy, and the daughter could inherit O from both parents. If O were dominant, the daughter could also be O. We need data from many offspring to decide.

- No: 1
- Reason (both possibilities fit / need more data): 1

SECTION C

Q9. A pea plant with violet flowers (Vv) is crossed with a plant with white flowers (vv). **[3]**

(i) Vv parent: V and v. vv parent: v only. [1]
(ii) Vv and vv in the ratio 1 : 1. [1]
(iii) 1 violet : 1 white [1]

Q10. How do Mendel's experiments show that (i) traits may be dominant or recessive, and (ii) traits are inherited independently? **[3]**

Answer: (i) In a monohybrid cross, all F_1 plants were tall, so tallness is dominant. Shortness reappeared in one-fourth of the F_2 plants, so it is recessive: it was hidden, not lost. (ii) In a dihybrid cross (round yellow × wrinkled green), new combinations such as round green and wrinkled yellow appeared in F_2. This shows seed shape and seed colour are inherited independently.

- Dominant/recessive from monohybrid: 1½
- Independent inheritance from dihybrid: 1½

OR, answer: Germ cells (gametes) are special cells that have only one set of chromosomes, half the number in body cells. When a male and a female germ cell fuse during fertilisation, the zygote gets the full number again. This keeps the chromosome number constant in every generation.

- Germ cells have half the number: 1
- Fusion restores full number: 1
- Constant across generations: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Parents: RRYY × rryy. Gametes: RY and ry. F_1: all RrYy, round and yellow. F_1 plants form four kinds of gametes: RY, Ry, rY and ry. In F_2, a 16-box checkerboard gives 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green. [3]

- Parents/gametes/F_1: 1
- F_1 gametes and checkerboard: 1
- F_2 ratio: 1

(ii) Round green and wrinkled yellow, which were not present in either parent. They prove that seed shape and seed colour are inherited independently of each other. [2]

- New combinations: 1
- Independent inheritance: 1

OR, answer: (a) (i) Humans: genetically. The father's sperm carries X or Y; XX gives a girl and XY gives a boy. (ii) In some reptiles, the temperature at which the fertilised eggs are kept decides whether the young will be male or female. (iii) Snails can change sex, which shows sex is not genetically fixed in them. (b) Yes. The mother always gives X, while the father gives X or Y with equal probability. Which sperm fertilises the egg is a matter of chance, so there is a 50% chance of either.

- (a) 1 each
- (b) 2

SECTION E

Q12. Earlobes can be free or attached. Assume that free earlobes (F) are dominant over attached earlobes (f). Both of Arjun's parents have free earlobes,… **[4]**

(i) ff [1]

(ii) Both are Ff (heterozygous). Each gave Arjun one f. [1]

(iii) Ff × Ff gives FF, Ff, Ff and ff, in the ratio 1 : 2 : 1. Only ff has attached earlobes, so the probability is 1/4 (25%). [2]

- Cross: 1
- 1/4: 1